EXPERIMENT : 1	ROLL NO	Installation of Hadoop and Moniter Hadoop Setup using Web-based Toll
DATE:	23DI16	

Aim:

To install Hadoop and Monitor setup using web-based tool

COMPONENTS REQUIRED:

S.no	FACILITIES REQUIRED	QUANTITY
1	System	01
2	Operating System	Ubuntu

Procedure:

Step 1: Install JDK on Ubuntu

- To install **JDK** , first update the ubuntu by

$~ sudo apt update

```
dit@dit-OptiPlex-3060:~$ sudo apt update
[sudo] password for dit:
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2,820 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy-updates/main i386 Packages [916 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3,084 kB]
Get:8 http://security.ubuntu.com/ubuntu jammy-security/main i386 Packages [733 kB]
Get:9 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [408 kB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/main amd64 DEP-11 Metadata [54.5 kB]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/main amd64 c-n-f Metadata [13.9 kB]
Get:12 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [4,726 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [476 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 DEP-11 Metadata [112 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 c-n-f Metadata [19.0 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 DEP-11 Metadata [212 B]
Get:17 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe i386 Packages [788 kB]
Get:18 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1,240 kB]
Get:19 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [308 kB]
Get:20 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 DEP-11 Metadata [359 kB]
Get:21 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [29.8 kB]
Get:22 http://in.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 DEP-11 Metadata [940 B]
Get:23 http://in.archive.ubuntu.com/ubuntu jammy-backports/main amd64 DEP-11 Metadata [7,152 B]
Get:24 http://in.archive.ubuntu.com/ubuntu jammy-backports/restricted amd64 DEP-11 Metadata [212
B]
Get:25 http://in.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 DEP-11 Metadata [9,696
B]
Get:26 http://in.archive.ubuntu.com/ubuntu jammy-backports/multiverse amd64 DEP-11 Metadata [212
```

- Type the following command in your terminal to install OpenJDK 8

$~ sudo apt install openjdk-8-jdk -y

```
kalai@kalai-Dit:~$ sudo apt install openjdk-8-jdk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java
  libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsm-dev
  libx11-dev libxau-dev libxcb1-dev libxdmcp-dev libxt-dev
  openjdk-8-jdk-headless openjdk-8-jre openjdk-8-jre-headless x11proto-dev
  xorg-sgml-doctools xtrans-dev
Suggested packages:
  default-jre libice-doc libsm-doc libx11-doc libxcb-doc libxt-doc
  openjdk-8-demo openjdk-8-source visualvm fonts-nanum fonts-ipafont-gothic
  fonts-ipafont-mincho fonts-wqy-microhei fonts-wqy-zenhei fonts-indic
The following NEW packages will be installed:
  ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java
  libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsm-dev
  libx11-dev libxau-dev libxcb1-dev libxdmcp-dev libxt-dev openjdk-8-jdk
  openjdk-8-jdk-headless openjdk-8-jre openjdk-8-jre-headless x11proto-dev
  xorg-sgml-doctools xtrans-dev
```

- Once the installation process is complete, verify the current Java version

Step 2: Install OpenSSH on Ubuntu

```
dit@dit-OptiPlex-3060:~$ java -version; javac -version
openjdk version "1.8.0_472"
OpenJDK Runtime Environment (build 1.8.0_472-8u472-ga-1~22.04-b08)
OpenJDK 64-Bit Server VM (build 25.472-b08, mixed mode)
javac 1.8.0_472
```

- Install the OpenSSH server and client using the following command

$~ sudo apt install openssh-server openssh-client -y

Step 3: Create Hadoop User

- Utilize the **adduser command** to create a new Hadoop user

$~ sudo adduser hdoop

- Switch to the newly created user and enter the corresponding password

```
dit@dit-OptiPlex-3060:~$ sudo adduser hdoop
Adding user `hdoop' ...
Adding new group `hdoop' (1001) ...
Adding new user `hdoop' (1001) with group `hdoop' ...
Creating home directory `/home/hdoop' ...
Copying files from `/etc/skel' ...
New password:
```

$~ su – hdoop

```
dit@dit-OptiPlex-3060:~$ su - hdoop
Password:
```

Step 4:Download Hadoop Package

- Use wget to download Hadoop:

$~ wget https://dlcdn.apache.org/hadoop/common/hadoop-3.4.0/hadoop-3.4.0.tar.gz

- Extract Hadoop:

$~ tar xzf hadoop-3.4.0.tar.gz

```
hdoop@kalai-Dit:~$ wget https://dlcdn.apache.org/hadoop/common/hadoop-3.4.0/hadoop-3.4.0.tar.gz
--2025-12-04 21:58:16--  https://dlcdn.apache.org/hadoop/common/hadoop-3.4.0/hadoop-3.4.0.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 965537117 (921M) [application/x-gzip]
Saving to: 'hadoop-3.4.0.tar.gz'

hadoop-3.4.0.tar.gz          100%[===================================================>] 920.81M  9.47MB/s    in 2m 22s

2025-12-04 22:00:38 (6.49 MB/s) - 'hadoop-3.4.0.tar.gz' saved [965537117/965537117]

hdoop@kalai-Dit:~$ tar xzf hadoop-3.4.0.tar.gz
hdoop@kalai-Dit:~$
```

Step 5: Configure Hadoop Environment Variables (.bashrc)

- Edit the .bashrc file:

$~ nano .bashrc

#Hadoop Related Options
export HADOOP_HOME=/home/hdoop/hadoop-3.4.0
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"

```
  GNU nano 7.2                                          .bashrc
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi
#Hadoop Related Options
export HADOOP_HOME=/home/hdoop/hadoop-3.4.0
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

- Run the following command after saving the .bashrc file:

$~ source ~/.bashrc

Step 6: Edit hadoop-env.sh File

export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64

```
# The java implementation to use. By default, this environment
# variable is REQUIRED on ALL platforms except OS X!
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64

# The language environment in which Hadoop runs. Use the Engli
# environment to ensure that logs are printed as expected.
export LANG=en_US.UTF-8
```

Step 7: Edit core-site.xml File

```
<configuration>
 <property>
  <name>hadoop.tmp.dir</name>
  <value>/home/hdoop/tmpdata</value>
 </property>
 <property>
  <name>fs.default.name</name>
  <value>hdfs://127.0.0.1:9000</value>
 </property>
</configuration>
```

```
  GNU nano 7.2                    /home/hdoop/hadoop-3.4.0/etc/hadoop/core-site.xml
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
  <name>hadoop.tmp.dir</name>
  <value>/home/hdoop/tmpdata</value>
</property>
<property>
  <name>fs.default.name</name>
  <value>hdfs://127.0.0.1:9000</value>
</property>
</configuration>
```

Step 8: Edit hdfs-site.xml File

```
<configuration>
 <property>
  <name>dfs.data.dir</name>
  <value>/home/hdoop/dfsdata/namenode</value>
 </property>
 <property>
  <name>dfs.data.dir</name>
  <value>/home/hdoop/dfsdata/datanode</value>
 </property>
 <property>
  <name>dfs.replication</name>
```

```
  <value>1</value>
 </property>
</configuration>
```

```
  GNU nano 7.2                    /home/hdoop/hadoop-3.4.0/etc/hadoop/hdfs-site.xml *
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
  <name>dfs.data.dir</name>
  <value>/home/hdoop/dfsdata/namenode</value>
</property>
<property>
  <name>dfs.data.dir</name>
  <value>/home/hdoop/dfsdata/datanode</value>
</property>
<property>
  <name>dfs.replication</name>
  <value>1</value>
</property>
</configuration>
```

Step 9 : Edit mapred-site.xml File

```
<configuration>
 <property>
  <name>mapreduce.framework.name</name>
  <value>yarn</value>
 </property>
</configuration>
```

```
<!--
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  you may not use this file except in compliance with the License.
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    http://www.apache.org/licenses/LICENSE-2.0

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  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
  <name>mapreduce.framework.name</name>
  <value>yarn</value>
</property>
</configuration>
```

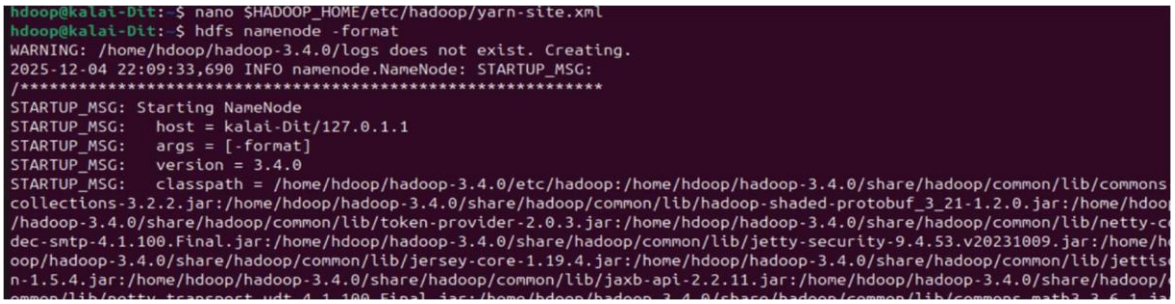
Step 10: Edit yarn-site.xml File

```
<configuration>
 <property>
  <name>yarn.nodemanager.aux-services</name>
  <value>mapreduce_shuffle</value>
 </property>
 <property>
  <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
  <value>org.apache.hadoop.mapred.ShuffleHandler</value>
 </property>
 <property>
  <name>yarn.resourcemanager.hostname</name>
  <value>127.0.0.1</value>
 </property>
 <property>
  <name>yarn.acl.enable</name>
  <value>0</value>
 </property>
 <property>
  <name>yarn.nodemanager.env-whitelist</name>

<value>JAVA_HOME,HADOOP_COMMON_HOME,HADOOP_HDFS_HOME,HADOOP_CONF
_DIR,CLASSPATH_PERPEND_DISTCACHE,HADOOP_YARN_HOME,HADOOP_MAPRED_H
OME</value>
 </property>
</configuration>
```

Step 11: Format HDFS NameNode

$~ hdfs namenode -format



Step 12: Start Hadoop Cluster

$~ ./start-dfs.sh

```
hdoop@kalai-Dit:~$ cd hadoop-3.4.0/sbin
hdoop@kalai-Dit:~/hadoop-3.4.0/sbin$ ./start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [kalai-Dit]
kalai-Dit: Warning: Permanently added 'kalai-dit' (ED25519) to the list of known hosts.
hdoop@kalai-Dit:~/hadoop-3.4.0/sbin$
```

$~. /start-yarn.sh

```
hdoop@kalai-Dit:~/hadoop-3.4.0/sbin$ ./start-yarn.sh
Starting resourcemanager
Starting nodemanagers
hdoop@kalai-Dit:~/hadoop-3.4.0/sbin$
```

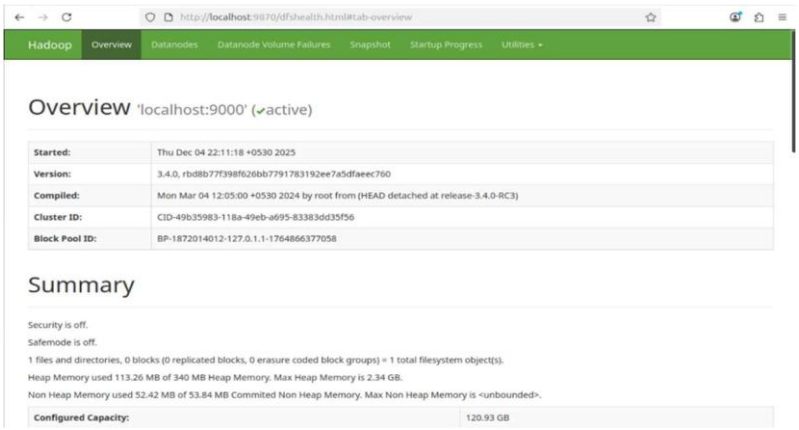
$~ jps

```
hdoop@kalai-Dit:~/hadoop-3.4.0/sbin$ jps
9617 NameNode
10727 Jps
10361 NodeManager
10235 ResourceManager
9966 SecondaryNameNode
9758 DataNode
hdoop@kalai-Dit:~/hadoop-3.4.0/sbin$
```

Step 13: Monitoring the Web-Tolls

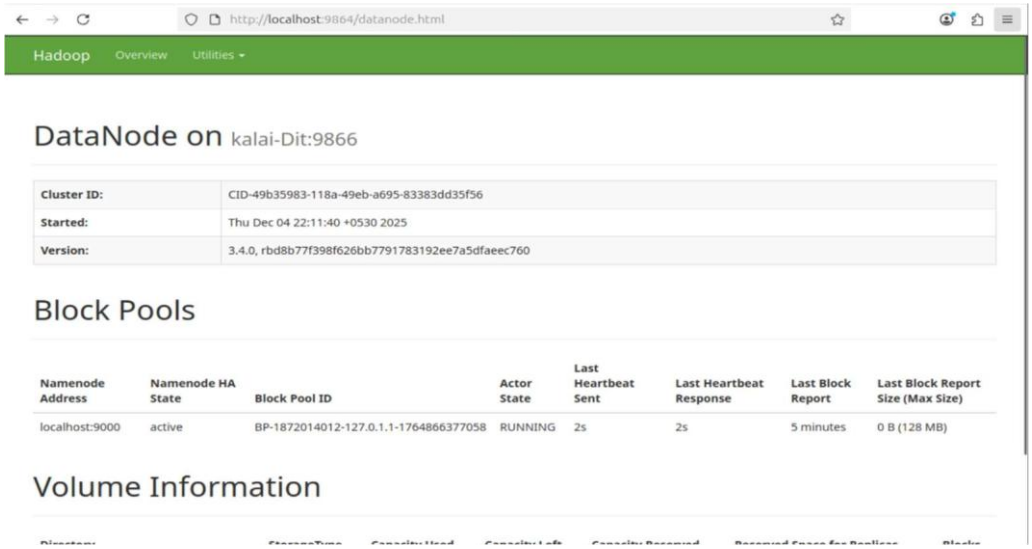
- Use the Localhosty ports to access the web tools
- Type the following url in your Browser (for overview)

https://localhost:9870



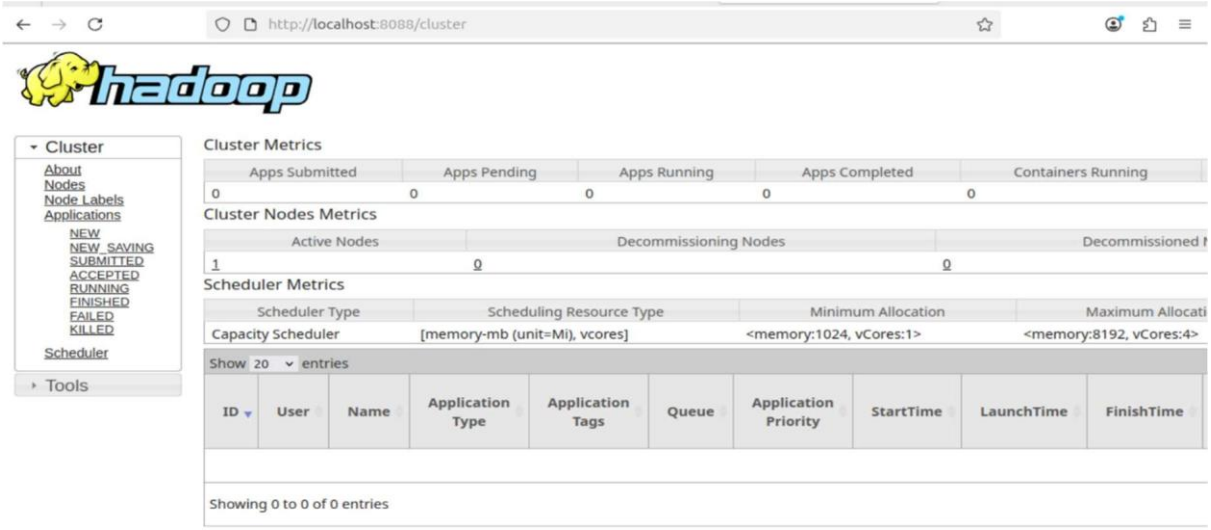
- Type the following url in your Browser (for Datanode)

https://localhost:9864



- Type the following url in your Browser (for Hadoop Cluster)

https://localhost:8088



Result :

Thus we have Successfully installed Hadoop and Monitor setup using web-based tool.